	Case Name: **Roadworks Poperinge**	Sector	**Construction (civil)**
	OR-AS Operations Research - Applications and Solutions www.or-as.be info@or-as.be	Baseline Schedule	Schedule with resources
			Schedule with costs
		Risk Analysis	Random simulation
			One of nine std. scenarios
Submitted by	**Matilde Casado and Margarida Antunes**		User defined distributions
Date	**2015**	Project Control	Automatic tracking
File Name	**C2015-34 Roadworks Poperinge**		Tracking based on user input

1. Project description

Project authenticity

The Roadworks Poperinge project includes demolition, excavation, sewer installation, foundation work, paving with concrete street stones, and finishing touches.

The project consists of activity, resource and cost data that were obtained directly from the actual project owner.

2. Project properties

2.1. Baseline Schedule

General	
# Activities	13
Planned Duration (PD)	120 days*
Budget At Completion (BAC)	511 325.9 €
Renewable Resources	-
Consumable Resources	-

* standard eight-hour working days

Network topology	
Serial/Parallel (SP)	91%
Activity Distribution (AD)	99%
Length of Arcs (LA)	0%
Topological Float (TF)	18%

2.2. Risk Analysis

Random simulation by ProTrack was performed using the default symmetric triangular risk distribution profiles.

	Cost sensitivity		
	avg [%]	std dev [%]	skew [-]
CRI-r	23.4	19.7	-0.8
CRI-rho	23.7	19.7	-0.9
CRI-tau	15.5	13.6	-1.0

	Resource sensitivity		
	avg [%]	std dev [%]	skew [-]
CRI-r	N/A	N/A	N/A
CRI-rho	N/A	N/A	N/A
CRI-tau	N/A	N/A	N/A

	Time sensitivity		
	avg [%]	std dev [%]	skew [-]
CI	65.5	41.6	0.5
SI	40.0	39.6	-0.5
SSI	35.9	30.8	-0.2
CRI-r	18.7	22.7	-2.0
CRI-rho	18.8	22.5	-2.1
CRI-tau	13.7	15.7	-2.3

2.3. Project Control

2.3.1. Simulated forecasting accuracy

The accuracy of time and cost forecasting methods has been evaluated based on Monte Carlo simulation runs using the risk profiles described in section "2.2. Risk Analysis". Based on these risk profiles, the Mean Absolute Percentage Error (MAPE) and Mean Percentage Error (MPE) has been calculated to evaluate the expected accuracy of the time and cost predictions, EAC(t) and EAC, respectively.

Simulated EAC(t) accuracy		
method - PF	MAPE [%]	MPE [%]
PV - 1	7.1	0.5
PV - SPI	9.5	0.6
PV - SCI	17.7	0.0
ED - 1	6.7	0.3
ED - SPI	9.2	0.3
ED - SCI	13.6	-0.2
ES - 1	6.2	0.2
ES - SPI(t)	8.0	-0.3
ES - SCI(t)	12.5	-0.5

Simulated EAC accuracy		
method (PF)	MAPE [%]	MPE [%]
1	3.7	0.5
CPI	6.1	1.4
SPI	5.2	0.8
SPI(t)	4.6	0.4
SCI	9.0	0.8
SCI(t)	8.4	0.6
0.8 CPI + 0.2 SPI	5.8	1.4
0.8 CPI + 0.2 SPI(t)	5.6	1.3

According to the MAPE values[1] the best performance for time forecasting can be expected from the unweighted Earned Schedule method. For cost forecasting the unweighted and CPI-weighted methods should yield the best results.

2.3.2. Tracking description

Tracking authenticity

Manual tracking was performed over 7 tracking periods with a length of approximately 2 months. The Real Duration and Real Cost mentioned in section "2.3.3. Earned Value Management" are based on manual user input.

The tracking information obtained from the project owner and introduced in ProTrack includes actual activity start dates, durations and costs.

[1] The MAPE gives the best indication for the forecast accuracy (the lower the MAPE, the more accurate the method) since all deviations from the targeted real duration (real cost) are cumulated, whereas for the MPE underestimates can be compensated by overestimates and vice versa, possibly leading to an overly positive evaluation of a certain method. However, the MPE can provide useful information about the nature of the deviations, i.e. does the method rather underestimate or overestimate the real duration (real cost)?

2.3.3. Earned Value Management

2.3.3.1. Performance metrics

	CV [€]	SV [€]	SV(t) [d]	CPI [-]	SPI [-]	SPI(t) [-]	p-factor [-]
avg	-3360.5	-173775.9	-597.9	1.0	0.6	0.3	0.8
std dev	56046.4	110171.1	206.2	0.1	0.3	0.2	0.1
final	70931.7	0.0	-943.0	1.2	0.9	0.5	1.0

2.3.3.2. Time forecasting

PD	120 days	Real Duration	238 days	Late	98.3%

EAC(t)			**Real Accuracy**	
method - PF	avg [d]	std dev [d]	MAPE [%]	MPE [%]
PV - 1	228.6	1.6	2.5	2.5
PV - SPI	229.3	1.4	2.5	2.5
PV - SCI	230.0	1.3	2.5	2.5
ED - 1	230.7	1.4	2.4	2.4
ED - SPI	231.4	1.6	2.4	2.4
ED - SCI	232.1	2.0	2.4	2.4
ES - 1	232.9	2.0	2.3	2.3
ES - SPI(t)	233.6	1.6	2.3	2.3
ES - SCI(t)	234.3	1.4	2.2	2.2

2.3.3.3. Cost forecasting

BAC	511 325.9 €	Real Cost	440 394.2 €	Under Budget	-13.9

EAC			**Real Accuracy**	
method (PF)	avg [€]	std dev [€]	MAPE [%]	MPE [%]
1	514686.3	56046.4	2.4	-2.4
CPI	529632.6	83216.2	2.9	-2.9
SPI	1384751.4	1766655.2	30.6	-30.6
SPI(t)	1468480.6	1248746.1	33.3	-33.3
SCI	1387105.7	1669916.0	30.7	-30.7
SCI(t)	1506102.9	1179144.8	34.6	-34.6
0.8 CPI + 0.2 SPI	562378.0	89455.7	4.0	-4.0
0.8 CPI + 0.2 SPI(t)	572970.7	96594.4	4.3	-4.3